Artificial Intelligence and Machine Learning - Answer Sheet - Student 12

Question 1: Explain the differences between supervised, unsupervised, and reinforcement learning in machine learning. Provide examples of applications for each approach.

Answer: So basically there are three main types of machine learning and they're all different in how they work. Supervised learning is when you have data that's already labeled which means someone has already told the computer what the right answer is. Like if you're trying to teach a computer to recognize cats in pictures, you would show it thousands of pictures and for each picture you would say "this is a cat" or "this is not a cat." Then the computer learns patterns from these examples. Examples of supervised learning include spam detection where emails are labeled as spam or not spam, and also medical diagnosis where doctors have already identified diseases in medical scans. Credit scoring is another example where banks know which customers defaulted on loans in the past. Weather prediction also uses supervised learning because we have historical weather data with known outcomes. Unsupervised learning is different because there are no labels. The computer has to figure out patterns by itself without being told what's right or wrong. It's like giving someone a box of mixed puzzle pieces from different puzzles and asking them to sort them into groups without knowing what the final pictures should look like. Clustering is a big part of unsupervised learning where similar things get grouped together. Customer segmentation is a common application where companies group customers based on buying habits without knowing ahead of time what the groups should be. Anomaly detection is another use case where the system learns what normal behavior looks like and then flags anything unusual. Topic modeling in text analysis is also unsupervised where the algorithm finds themes in documents without being told what topics to look for. Reinforcement learning is like training a pet or teaching a child through rewards and punishments. The computer agent tries different actions and gets feedback about whether those actions were good or bad. Over time it learns which actions lead to better outcomes. It's trial and error learning basically. Game playing is the most famous example like AlphaGo beating human champions at Go. Self-driving cars also use reinforcement learning to learn how to navigate traffic safely. Robots learning to walk or manipulate objects use this approach too. Trading algorithms in finance can use reinforcement learning to make buy and sell decisions.

Question 2: Describe the architecture and functioning of Convolutional Neural Networks(CNNs) and explain why they are particularly effective for image recognition tasks.

Answer: CNNs are really cool neural networks that are specially designed for images. They have these things called convolutional layers which are the main part that makes them work so well. What happens is these layers apply filters to the image, and these filters are like little templates that look for specific features. For example, one filter might look for horizontal edges, another for vertical edges, and so on. The way convolution works is the filter slides across the entire image, and at each position it does a mathematical operation called convolution which is basically multiplying the filter values with the pixel values underneath and adding them up. This creates what's called a feature map that shows where that particular feature appears in the image. After the convolutional layers there are usually activation functions like ReLU which just sets negative values to zero. This adds non-linearity which is important for the network to learn complex patterns. Then there are pooling layers, usually max pooling, which take the maximum value in small regions of the feature map. This makes the representation smaller and also provides some translation invariance. The reason CNNs work so well for images is because they understand the spatial structure of images. In a regular neural network, if you moved an object from one part of the image to another, the network might not recognize it as the same object. But CNNs use the same filters across the whole image, so they can detect features regardless of where they appear. Also, the hierarchical structure is really important. Early layers detect simple features like edges and corners. Middle layers combine these simple features to detect more complex patterns like textures and shapes. Deep layers combine these to recognize complete objects like faces or cars. It's kind of like how human vision works, starting with simple features and building up to complex understanding. Parameter sharing is another big advantage because the same filter is used across the entire image, which means CNNs need way fewer parameters than fully connected networks. This makes them more

efficient and less likely to overfit.

Question 3: Discuss the ethical considerations and potential societal impacts of implementing artificial intelligence systems in critical decision-making processes.

Answer: This is a really important topic because AI is being used more and more in situations that really affect people's lives. One of the biggest problems is bias. AI systems learn from data, and if that data contains biases from society, then the AI will learn those biases too. For example, if a hiring system is trained on data from a company thathistorically hired mostly men for engineering positions, the AI might learn that being male is somehow important for engineering jobs, even though that's obviously not true. This kind of bias can show up in lots of different areas. Criminal justice systems that predict recidivism rates have been shown to be biased against certain racial groups. Facial recognition systems work better on white males than on women or people of color. Loan approval systems might discriminate against people from certain neighborhoods or backgrounds. Another big issue is transparency and explainability. Many AI systems, especially deep learning models, are like black boxes where you can't easily understand how they make decisions. This is a problem when the decisions are important, like medical diagnoses or legal judgments. Doctors need to understand why an AI recommended a particular treatment, and judges need to understand why an AI suggested a particular sentence. Privacy is also a huge concern. AI systems often need lots of personal data to work effectively. This raises questions about who owns this data, how it's being used, and whether people have given proper consent. There's also the risk that AI systems can infer sensitive information about people from data that seems harmless. For example, an AI might be able to predict someone's sexual orientation from their social media likes, even if they never explicitly shared that information. Job displacement is another major societal impact. AI and automation are likely to eliminate many jobs, and while new jobs might be created, there could be a difficult transition period. Some people might not be able to retrain for new roles, leading to unemployment and economic inequality. There's also the question of accountability. When an AI system makes a mistake that harms someone, who is responsible? Is it the company that made the AI, the person who deployed it, or the person who used it? This is especially complicated because AI development often involves many different parties.

Question 4: Explain the concept of transfer learning in deep neural networks and discuss its advantages and limitations.

Answer: Transfer learning is this really clever technique where instead of training a neural network from scratch, you start with a network that's already been trained on a different but related task. It's like if you already knew how to play the piano and then wanted to learn the organ - you wouldn't start from zero because many of the skills transfer over. The way it typically works is you take a neural network that's been pre-trained on a huge dataset, like ImageNet which has millions of images across thousands of categories. This network has already learned to recognize lots of basic visual featureslike edges, shapes, textures, and even some complex patterns. Then you adapt this network for your specific task. There are different ways to do transfer learning. One approach is feature extraction where you freeze most of the pre-trained network and only train the final layers for your specific task. Another approach is fine-tuning where you take the pre-trained weights as a starting point but then continue training the entire network on your data, usually with a lower learning rate so you don't mess up the useful features that were already learned. The advantages are really significant. First, you need way less training data for your specific task because the network has already learned general features from the large pre-training dataset. This is especially helpful in domains like medical imaging where it's hard to get large labeled datasets. Second, training is much faster because you're not starting from random weights. Third, you often get better performance, especially when your dataset is small, because the pre-trained features provide a much better starting point than random initialization. Transfer learning also helps prevent overfitting because the pre-trained network has learned regularities from a large diverse dataset, and these regularities transfer to your smaller dataset. But there are limitations too. The biggest one is that transfer learning works best when the source and target tasks are similar. If you try to transfer from a network trained on natural images to one that needs to analyze medical scans or satellite imagery, the benefit might be limited. Sometimes you might even get negative transfer where the pre-trained features actually hurt performance on your task. Another limitation is that the

architecture of the pre-trained network might not be optimal for your specific task. You're kind of stuck with the design decisions that were made for the original task. Also, any biases that were present in the pre-training data can transfer to your application, which could be problematic in sensitive applications.

Question 5: Describe the principles of natural language processing (NLP) and how transformer-based models like BERT have revolutionized language understanding tasks.

Answer: NLP is all about getting computers to understand and work with human language, which is incredibly challenging because language is so complex and ambiguous. Traditional approaches used rule-based systems where linguists would manually encode grammar rules and language patterns, but this was really limited and couldn't handle the full complexity of natural language.Then statistical methods became popular where systems would learn patterns from large amounts of text data. N-gram models would predict the next word based on the previous few words. Bag-of-words models would represent documents as collections of words without considering order. These worked better than rule-based systems but still had major limitations, especially with understanding context and meaning. The introduction of neural networks improved things significantly. Recurrent neural networks like LSTMs could process sequences of words and maintain some memory of previous words, which helped with understanding context. But they still processed text sequentially, one word at a time, which made them slow to train and limited their ability to capture long-range dependencies. Then transformers came along and completely changed everything. The key innovation was the self-attention mechanism, which allows the model to look at all words in a sentence simultaneously and determine which words are most relevant to each other. Instead of processing words one by one, transformers can process entire sequences in parallel, making them much faster to train. BERT specifically was revolutionary because it introduced bidirectional training. Previous models like GPT only looked at words that came before the current word, but BERT looks at context from both directions. During training, BERT randomly masks some words in sentences and tries to predict what those words should be based on the surrounding context from both sides. This bidirectional training, combined with training on massive amounts of text data, allows BERT to develop really rich representations of words that capture their meaning in context. The same word can have different representations depending on how it's used in a sentence, which is crucial for understanding language. The pre-training and fine-tuning approach is also really powerful. BERT is first pre trained on general language understanding tasks using huge amounts of unlabeled text. Then it can be fine-tuned for specific tasks like sentiment analysis, question answering, or text classification with relatively small amounts of labeled data. This has led to huge improvements across pretty much all NLP benchmarks. Tasks that used to require very different approaches can now be solved by fine-tuning BERT with minimal changes to the architecture. But there are still challenges. These models require enormous computational resources to train. They can perpetuate biases present in their training data. And while they're very good at pattern matching and statistical relationships, there are questions about whether they truly understand language or are just very sophisticated pattern matchers.